

Squat Analyzer

AI-Powered Squat Form Analysis

*"Better form. Fewer injuries.
No coach required."*

33

body landmarks
detected in real-time

3

deep learning
models in production

3

Seconds from Video to Scoring

The Problem

Injury Risk

Poor form is a primary cause of knee & lower-back injuries in strength training

Coach Cost

Personal trainers cost \$60-120/session
— unaffordable for daily workouts

No Feedback Loop

Bad habits compound silently — users have no idea their form is degrading

Hardware Barriers

Existing solutions require expensive motion-capture gear or gym sensors

Our Solution

AI form coaching — just a webcam, no special hardware

01

CAPTURE

Live webcam, video upload, or static image
MediaPipe Tasks Vision detects 33
3D body landmarks in-browser, zero
latency

02

ANALYZE

FastAPI backend runs deep-learning
models, detects active rep, classifies
form quality

03

SCORE

Instant multi-dimensional feedback:
Good/Bad quality rating, session
score 0–4

How It Works

1 Pose Detection

MediaPipe detects
33 3D landmarks
per frame in-browser

2 Ugly Recording detection

Real-time rule-based detection of
recording quality

3 Activity Segment

Start/Stop
isolates active
squat frames

4 Exercise Classify

Predict probability of being a squat
exercise

5 Session Score

Regression model
returns 0.0–4.0 score
(0 = perfect form)

6 3D Replay

Interactive skeleton
viewer shows every
rep frame in 3D

Features at a Glance

Live Webcam Mode



Real-time frame-by-frame pose analysis with instant depth & quality feedback

Video & Image Upload



Analyze existing footage via drag-and-drop — supports video files and static images

3D Skeleton Replay



Interactive 3D viewer replays every frame of the session with joint overlay

Session Scoring



Quantified 0.0–4.0 score per session. Automated end-to-end in under 3 seconds

Good / Bad Classifier



Binary quality prediction with confidence score using inter-joint features

Data Export



Save sessions to Supabase database or download full 3D keypoints as CSV

Technology Stack

FRONTEND & DEPLOYMENT

React + Vite + Tailwind CSS

Modern SPA with fast HMR builds

MediaPipe Tasks Vision

In-browser 33-landmark pose detection

Supabase (PostgreSQL)

Keypoint session storage

Vercel

CDN-edge frontend deployment

GitHub Actions

CI/CD — lint, test, build on every PR

BACKEND & ML

FastAPI (Python)

REST API with OpenAPI docs, async endpoints

PyTorch

GRU, LSTM, CNN model inference at runtime

scikit-learn

SVM, ensemble models, feature pipelines

MLflow on DagsHub

Experiment tracking & model registry

Render

Auto-scaling backend web service hosting

Meet the Team

4 developers · Full-Stack · ML Research · DevOps



Samuel

Team Lead
Full-Stack & DevOps



Jesper

ML Engineer
ML Research & Development



Nasser

Full Stack Engineer
Frontend & Backend



Emil

Test Engineer
Testing & DevOps

What's Next

Mobile App



Mobile app for easy scoring on the go

Weakest Link



Point of failure for exercise

Multi-Exercise



Deadlift, bench press, lunge support